13.2: Fertilization and meiosis alternate in sexual life cycles

A **life cycle** is the generation-to-generation sequence of stages in the reproductive history of an organism, from conception to production of its own offspring. In this section, we use humans as an example to track the behavior of chromosomes through the sexual life cycle. We begin by considering the chromosome count in human somatic cells and gametes. We will then explore how the behavior of chromosomes relates to the human life cycle and other types of sexual life cycles.

Sets of Chromosomes in Human Cells

In humans, each somatic cell has 46 chromosomes. Before mitosis begins, the chromosomes are duplicated. During mitosis, the chromosomes become condensed enough to be visible under a light microscope. At this point, they can be distinguished from one another by their size, the position of their centromeres, and the pattern of colored bands produced by certain chromatin-binding stains.

Careful examination of a micrograph of the 46 human chromosomes from a single cell in mitosis reveals that there are two chromosomes of each of 23 types. This becomes clear when images of the chromosomes are arranged in pairs, starting with the longest chromosomes. The resulting ordered display is called a **karyotype(Figure 13.3)**. The two chromosomes of a pair have the same length, centromere position, and staining pattern: These are called **homologous chromosomes** (or **homologs**). Both chromosomes of each pair carry genes controlling the same inherited characters. For example, if a gene for eye color is situated at a particular locus on a certain chromosome, then its homologous chromosome (its homolog) will also have a version of the eye-color gene at the equivalent locus.

Figure 13.3

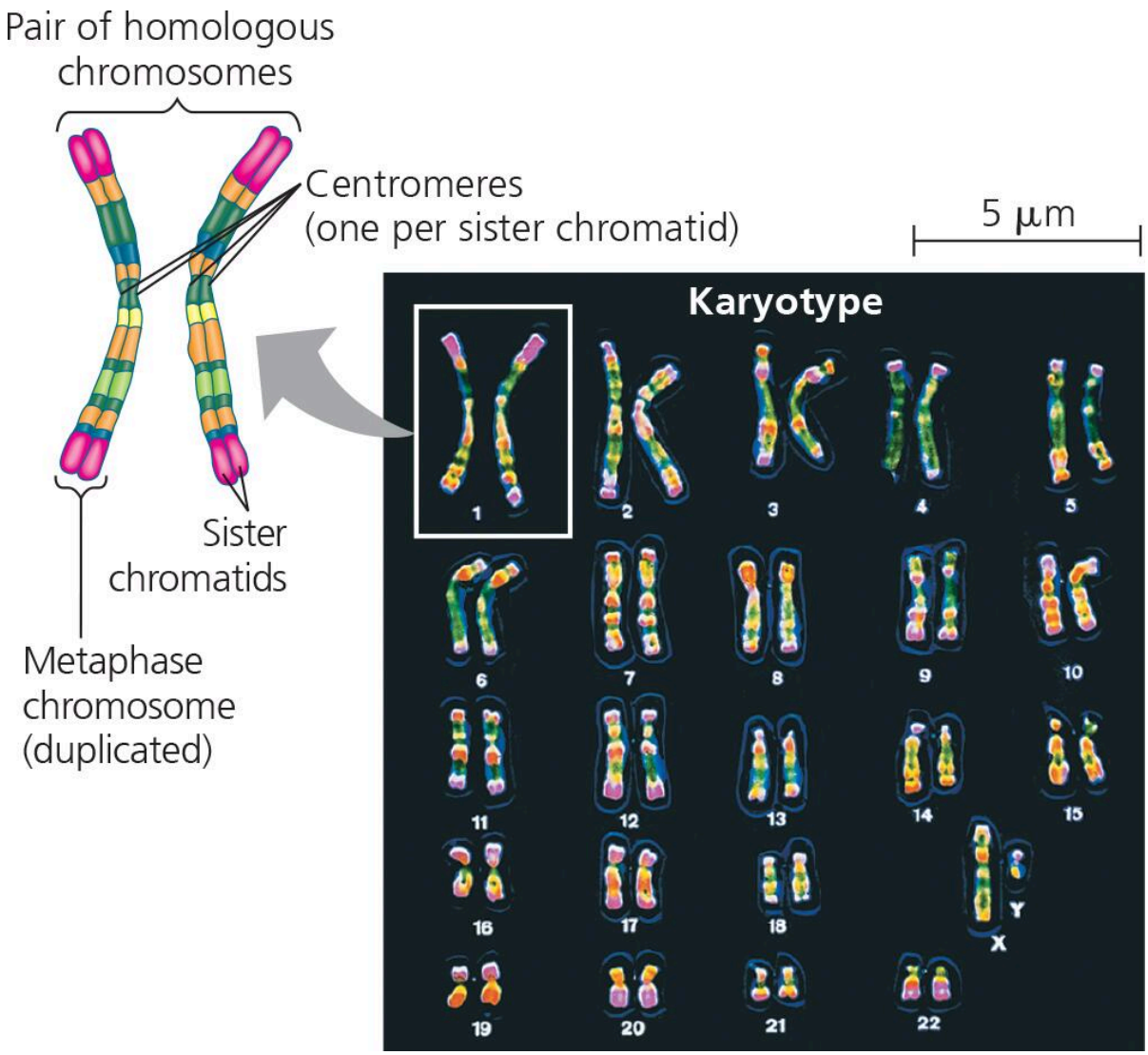
Research Method: Preparing a Karyotype

Application

A karyotype is a display of condensed chromosomes arranged in pairs. Karyotyping can be used to screen for defective chromosomes or abnormal numbers of chromosomes associated with certain congenital disorders, such as Down syndrome.

Technique

Karyotypes are prepared from isolated somatic cells, which are treated with a drug to stimulate mitosis and then grown in culture for several days. Cells arrested when the chromosomes are most highly condensed—at metaphase—are stained and then viewed with a microscope equipped with a digital camera. An image of the chromosomes is displayed on a computer monitor, and digital software is used to arrange them in pairs according to their appearance.



Human karyotype is an image of 23 pairs of homologous chromosomes, labeled 1 through 22 for somatic cells. Chromosomes clearly vary in length and color banding patterns, but chromosomes in each homologous pair show similar length and color banding patterns. X and Y chromosomes are visible after chromosome pair 22, indicating that the karyotype is from a human male. The X chromosome is larger than the Y chromosome and contains different color banding patterns.

The image zooms in on metaphase homologous pair 1, with a diagram of this pair at the side of the karyotype. The diagram shows this pair of homologous chromosomes in sister chromatid form, indicating that they have been duplicated and have similar centromere positions.

Results

This karyotype shows the chromosomes from a human male (as seen by the presence of the XY chromosome pair), colored to emphasize their chromosome banding patterns. The size of the chromosome, position of the centromere, and pattern of stained bands help identify specific chromosomes. Although difficult to discern

in the karyotype, each metaphase chromosome consists of two closely attached sister chromatids (see the diagram of the first pair of homologous duplicated chromosomes).

MAKE CONNECTIONS *Explain why each chromosome in the karyotype consists of two sister chromatids. (See**Figures 12.6**and*12.7.)*

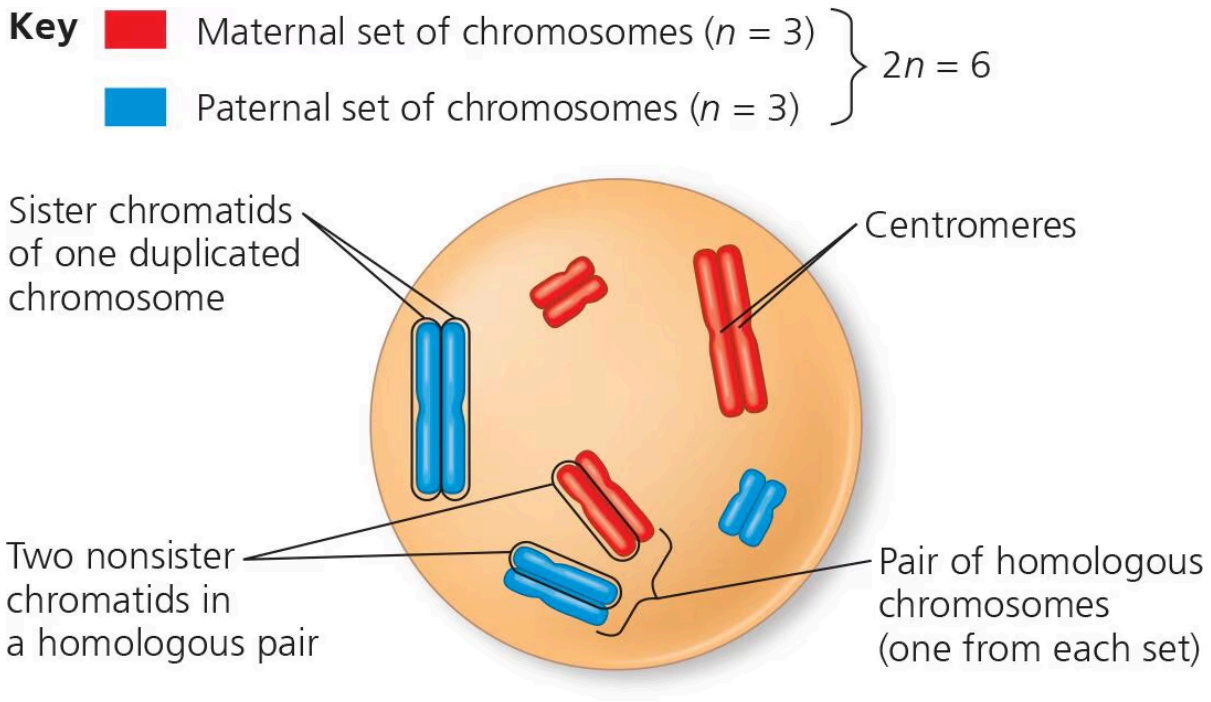
*For suggested answer, see*Appendix A.*

The two chromosomes referred to as X and Y are an important exception to the general pattern of homologous chromosomes in human somatic cells. Typically, human females have a homologous pair of X chromosomes (XX), while males have one X and one Y chromosome (XY; see Figure 13.3). Only small parts of the X and Y are homologous. Most of the genes carried on the X chromosome do not have counterparts on the tiny Y, and the Y chromosome has genes lacking on the X. Due to their role in sex determination, the X and Y chromosomes are called **sex chromosomes**. The other chromosomes are called **autosomes**.

The occurrence of pairs of homologous chromosomes in each human somatic cell is a consequence of our sexual origins. We inherit one chromosome of a pair from each parent. Thus, the 46 chromosomes in our somatic cells are actually two sets of 23 chromosomes—a maternal set (from our female parent) and a paternal set (from our male parent). The number of chromosomes in a single set is represented by n*.* Any cell with two chromosome sets is called a **diploid cell** and has a diploid number of chromosomes, abbreviated 2n. For humans, the diploid number is 46(2n=46), the number of chromosomes in our somatic cells. In a cell in which DNA synthesis has occurred, all the chromosomes are duplicated, and therefore each consists of two identical sister chromatids, associated closely at the centromere and along the arms. (Even though the chromosomes are duplicated, we still say the cell is diploid, or 2n. This is because it has only two sets of information regardless of the number of chromatids, which are merely copies of the information in one set.) **Figure 13.4** helps clarify the various terms that we use to describe duplicated chromosomes in a diploid cell.

Figure 13.4

Describing chromosomes.



Homologous chromosome sets are distinguished by color with blue homologs representing the paternal set of chromosomes and red homologs representing the maternal. Each homologous pair contains one maternal and one paternal homolog. Homologous chromosome pair #1 is long, homologous chromosome pair #2 is medium length, and homologous chromosome pair #3 is short. Here all six chromosomes are in duplicated sister chromatid form; sister chromatids are identified as being the same color and attached to each other at the centromere. Homologous chromosomes contain one blue set of sister chromatids and one red set for each of the long, medium and short pairs. Within a homologous pair (same length maternal and paternal chromosome sets), one blue sister chromatid would be a nonsister chromatid to a red chromatid. Nonsister chromatids are not attached to each other.

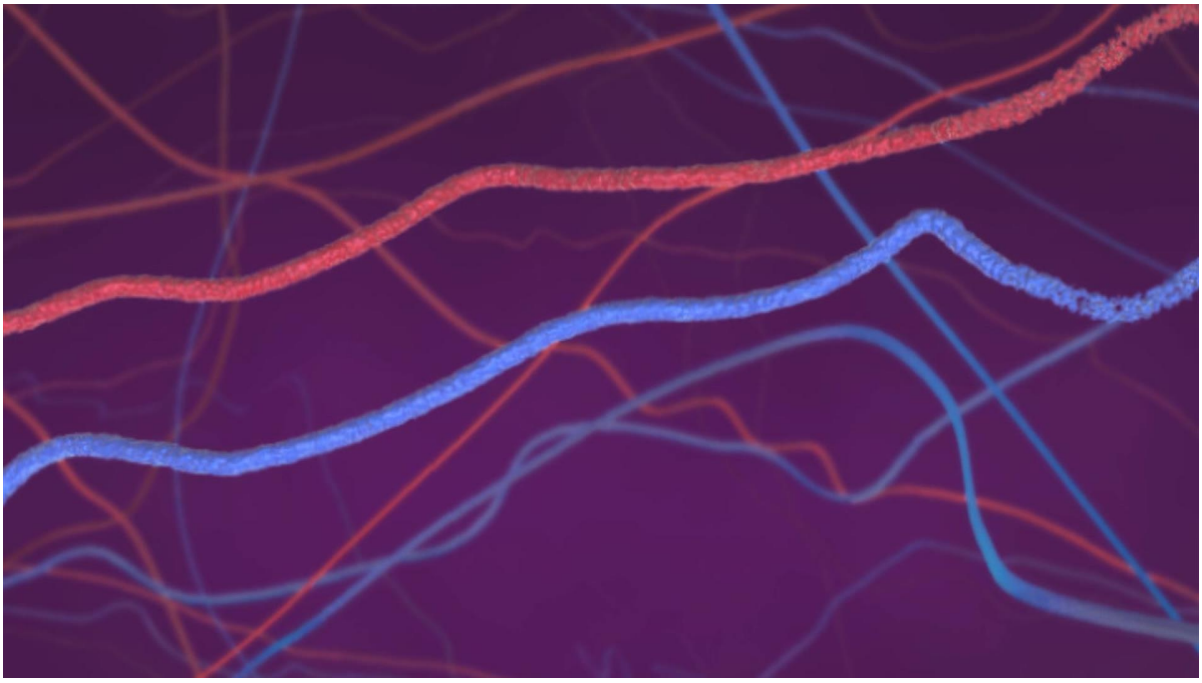
A cell from an organism with a diploid number of 62n=6 is depicted here following chromosome duplication and condensation. Each of the six duplicated chromosomes consists of two sister chromatids associated closely along their lengths. Each homologous pair is composed of one chromosome from the maternal set (red) and one from the paternal set (blue). Each set is made up of three chromosomes in this example (long, medium, and short). Together, one maternal and one paternal chromatid in a pair of homologous chromosomes are called nonsister chromatids.

VISUAL SKILLS *How many sets of chromosomes are present in this diagram? How many pairs of homologous chromosomes are present? If the phases of mitosis in* ***Figure 12.7*** *were redrawn using the color scheme shown in this figure, describe how the six chromosomes would be colored.*

*For suggested answer, see*Appendix A.*

BioFlix® Animation

Chromosomes



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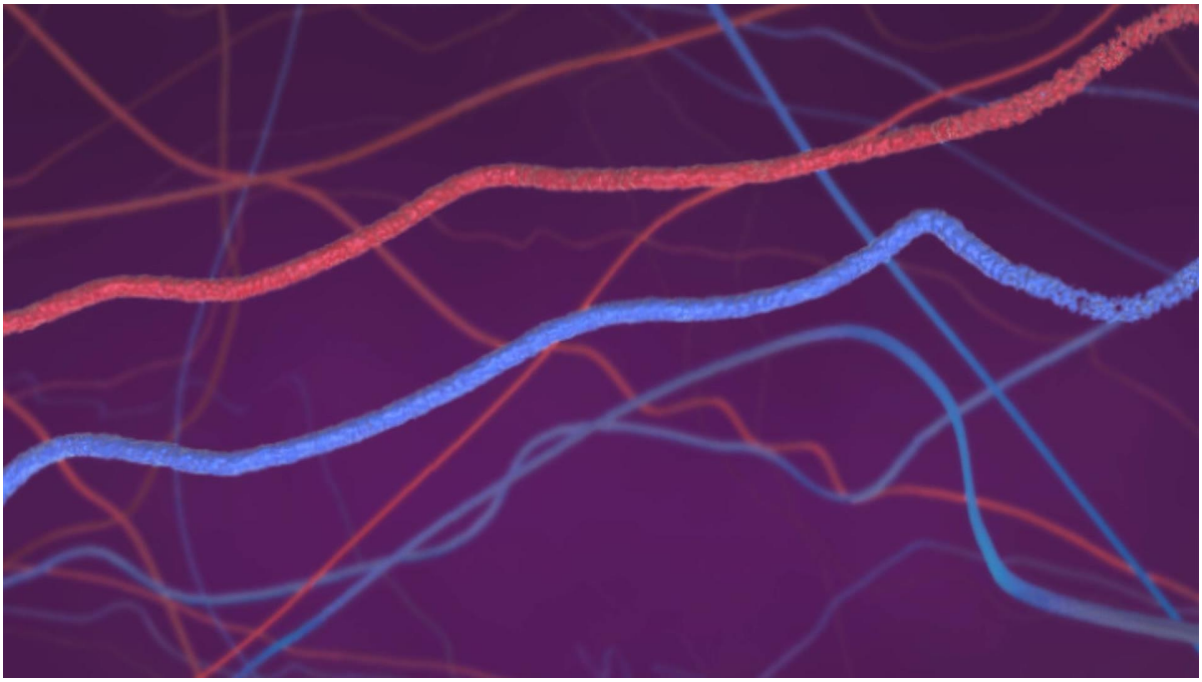

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Unlike somatic cells, gametes contain a single set of chromosomes. Such cells are called **haploid cells**, and each has a haploid number of chromosomes (n). For humans, the haploid number is 23(n=23). The set of 23 consists of the 22 autosomes plus a single sex chromosome. An unfertilized egg contains an X chromosome; a sperm contains either an X or a Y chromosome.

Each sexually reproducing species has a characteristic diploid and haploid number. For example, the fruit fly *Drosophila melanogaster* has a diploid number (2n) of 8 and a haploid number (n) of 4, while for dogs, 2n is 78 and n is 39. The chromosome number generally does not correlate with the size or complexity of a species' genome; it simply reflects how many linear pieces of DNA make up the genome, which is a function of the evolutionary history of that species (see Concept 21.5). Now let's consider chromosome behavior during sexual life cycles. We'll use the human life cycle as an example.

Behavior of Chromosome Sets in the Human Life Cycle

The human life cycle begins when a haploid sperm from the father fuses with a haploid egg from the mother **(Figure 13.5)**. This union of gametes, culminating in fusion of their nuclei, is called ********fertilization**. The resulting fertilized egg, or **zygote**, is diploid because it contains two haploid sets of chromosomes bearing genes representing the maternal and paternal family lines. As a human develops into a sexually mature adult, mitosis of the zygote and its descendant cells generates all the somatic cells of the body. Both chromosome sets in the zygote and all the genes they carry are passed with precision to the somatic cells.

Figure 13.5

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The human life cycle.
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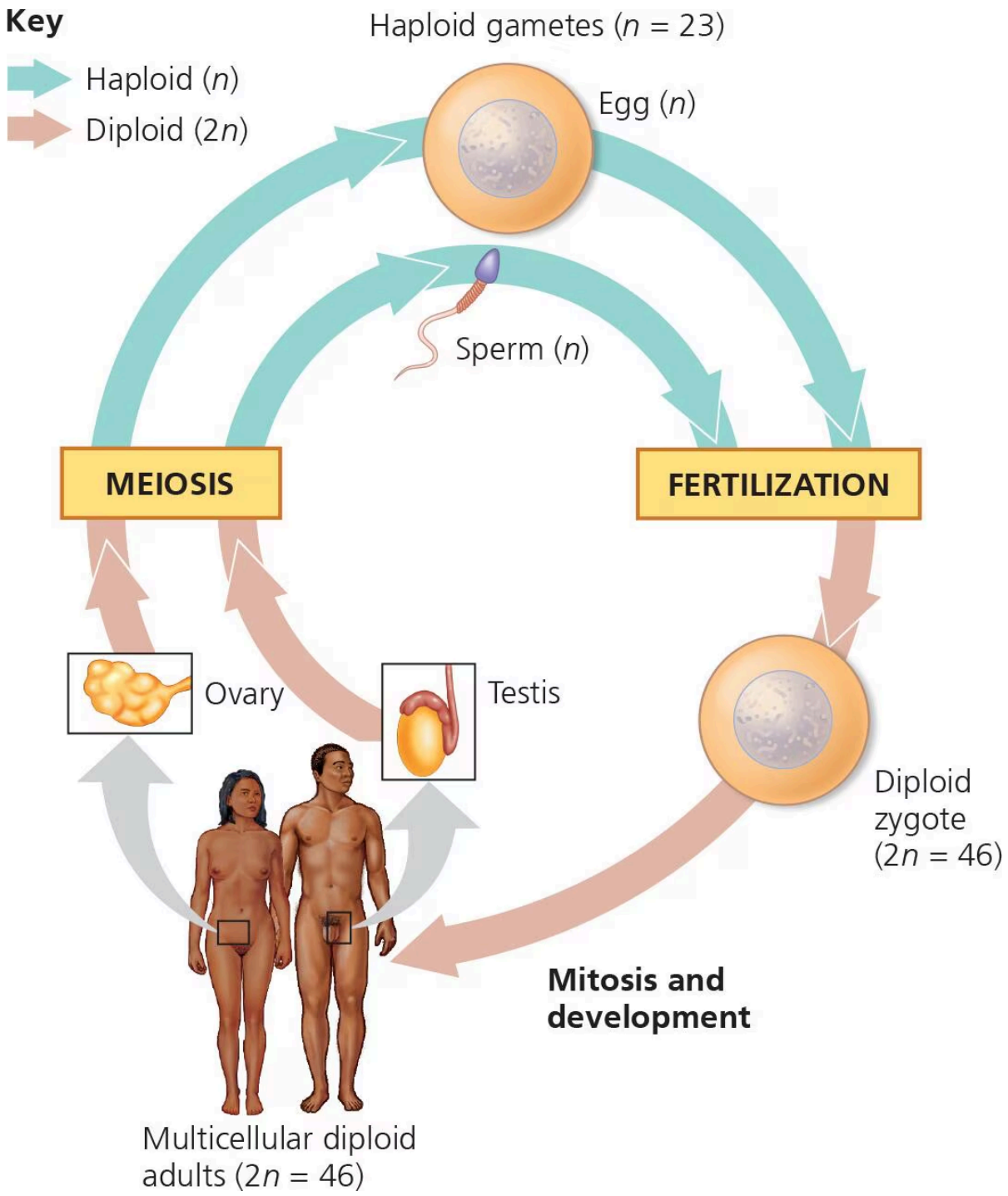


This figure introduces a color code that will be used for other life cycles later in this book. The aqua arrows identify haploid stages of a life cycle, and the tan arrows identify diploid stages.

Arrows depict the human life cycle containing both haploid and diploid stages.

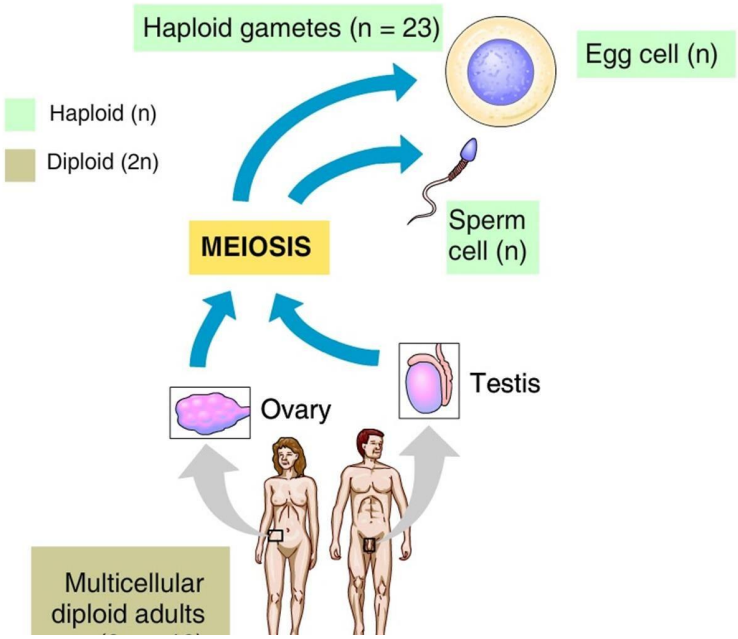
Multicellular diploid adults (a woman and a man) with 2 n equals to 46. Tan (diploid) arrows indicate meiosis occurs in diploid cells in the woman's ovary or the man's testes, with aqua arrows following meiosis indicating that a haploid stage has begun. The gametes resulting from meiosis (eggs for the woman and

sperm for the man) are both haploid (n equals to 23). Aqua (haploid) arrows flow to fertilization, where the gametes meet to form a diploid zygote (2 n equals to 46). Tan (diploid) arrows flow from the diploid zygote through the processes of mitosis and development back to the multicellular diploid adult stage (2 n equals to 46), at which point the cycle begins again.

In each generation, the number of chromosome sets per cell is halved during meiosis but doubles at fertilization. For humans, the number of chromosomes in a haploid cell is 23 consisting of one set n=23; the number of chromosomes in the diploid zygote and all somatic cells arising from it is 46, consisting of two sets 2n=46.

Animation

The Human Life Cycle



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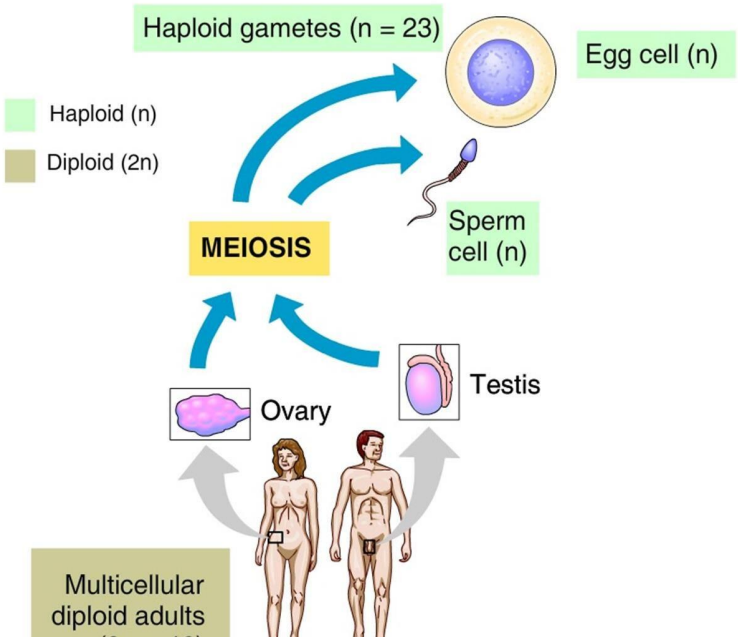

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The only cells of the human body not produced by mitosis are the gametes, which develop from specialized cells called *germ cells* in the gonads—ovaries in females and testes in males (see Figure 13.5). Imagine what would happen if human gametes were made by mitosis: They would be diploid like the somatic cells. At the next round of fertilization, when two gametes fused, the normal chromosome number of 46 would double to 92, and each subsequent generation would double the number of chromosomes yet again. This does not happen, however, because in sexually reproducing organisms, gamete formation involves a type of cell division called **meiosis**. This type of cell division reduces the number of sets of chromosomes from two in the parent cell to one in each gamete, counterbalancing the doubling that occurs at fertilization. As a result of meiosis, each human sperm and egg is haploid (n=23). Fertilization restores the diploid condition by combining two sets of chromosomes, and the human life cycle is repeated, generation after generation (see Figure 13.5).

The steps of the human life cycle are typical of many sexually reproducing animals. Indeed, fertilization and meiosis are also the hallmarks of sexual reproduction in plants, fungi, and protists just as in animals. Fertilization and meiosis alternate in sexual life cycles, maintaining a constant number of chromosomes in a species from one generation to the next.

The Variety of Sexual Life Cycles

Although the alternation of meiosis and fertilization is common to all organisms that reproduce sexually, the timing of these two events in the life cycle varies, depending on the species. These variations can be grouped into three main types of life cycles. In the type that occurs in humans and most other animals, gametes are the only haploid cells **(Figure 13.6a)**. Meiosis occurs in germ cells during the production of gametes, which undergo no further cell division prior to fertilization. After fertilization, the diploid zygote divides by mitosis, producing a multicellular organism that is diploid.

Figure 13.6

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Three types of sexual life cycles.
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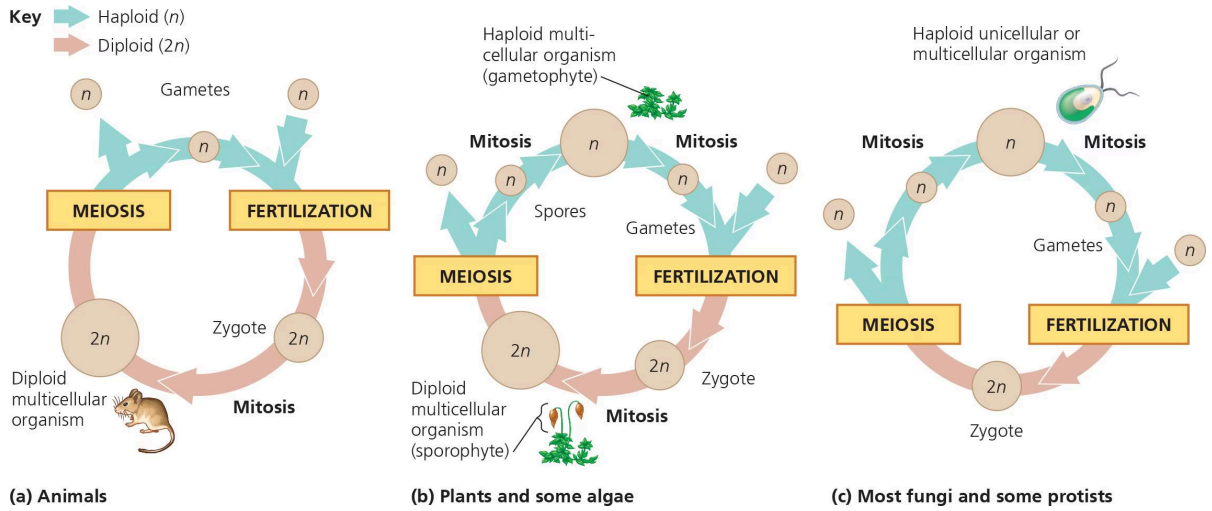

The common feature of all three cycles is the alternation of meiosis and fertilization, key events that contribute to genetic variation among offspring. The cycles differ in the timing of these two key events. (Small circles are cells; large circles are organisms.)

Types of cycles are as follows.

- Animals. Meiosis produces haploid gametes from cells in the diploid multicellular organism (mouse). Fertilization results from the fusion of haploid gametes; the resulting diploid zygote develops via mitosis. Diploid (2 n) multicellular organism (mouse) flows via diploid (tan) arrows to meiosis; haploid (aqua) arrows begin and flow to gametes (n). Fusion with another haploid gamete results in fertilization; diploid (tan) arrows flow to the zygote (2 n). Diploid arrows flow back to the multicellular mouse via mitosis and the cycle continues.
- b. Plants and some algae. The life cycle includes multicellular diploid sporophyte alternating with multicellular haploid gametophyte; haploid spores result from meiosis; gametes produced via

mitosis. Fusion of gametes equals fertilization. Cells in diploid (2 n) multicellular organisms (sporophyte) undergo meiosis, flowing by haploid arrows to produce the spores (n). Spores flow via haploid arrows by mitosis to form a haploid multicellular organism called the gametophyte. Haploid arrows flow from the gametophyte to produce the haploid gametes via mitosis. Fusion with another haploid gamete results in fertilization, after which diploid arrows flow to the diploid zygote (2 n). Diploid arrows flow from the zygote back to the sporophyte via mitosis and the cycle begins again.

- c. Most fungi and some protists. Meiosis produces haploid cells that develop via mitosis into haploid organism. From here, gametes produced via mitosis fuse via fertilization to produce diploid zygote. A haploid unicellular or multicellular organism produces haploid gametes via mitosis; fusion with another haploid gamete flows via haploid arrows to fertilization. Diploid arrows flow to diploid zygote (2 n), from which haploid cells are produced via meiosis. Some haploid cells flow via haploid arrows and the process of mitosis back to the haploid unicellular organism and the cycle begins again.

VISUAL SKILLS *For each type of life cycle, indicate whether haploid cells undergo mitosis, and if they do, describe the cells that are formed.*

*For suggested answer, see*Appendix A*.*

Plants and some species of algae exhibit a second type of life cycle called **alternation of generations(Figure 13.6b)**. This type includes both diploid and haploid stages that are multicellular. The multicellular diploid stage is called the *sporophyte*. Meiosis in the sporophyte produces haploid cells called *spores*. Unlike a gamete, a haploid spore doesn't fuse with another cell but divides mitotically, generating a multicellular haploid stage called the *gametophyte*. Cells of the gametophyte give rise to gametes by mitosis. Fusion of two haploid gametes at fertilization results in a diploid zygote, which develops into the next sporophyte generation. Therefore, in this type of life cycle, the sporophyte generation produces a gametophyte as its offspring, and the gametophyte generation produces a sporophyte. The name *alternation of generations* fits well for this type of life cycle.

A third type of life cycle occurs in most fungi and some protists, including some algae **(Figure 13.6c)**. After gametes fuse and form a diploid zygote, meiosis occurs without a multicellular diploid offspring developing. Meiosis produces not gametes but haploid cells that then divide by mitosis and give rise to either unicellular descendants or a haploid multicellular adult organism. Subsequently, the haploid organism carries out further mitoses, producing the cells that develop into gametes. The only diploid stage found in these species is the single-celled zygote.

Note that *either* haploid or diploid cells can divide by mitosis, depending on the type of life cycle. Only diploid cells, however, can undergo meiosis: Haploid cells can't because they already have a single set of chromosomes that cannot be further reduced. Though the three types of sexual life cycles differ in the timing of meiosis and fertilization, they share a fundamental result: genetic variation among offspring.